

---

# Is the Reconstruction–Forecasting Tradeoff a Frontier or an Operating Point?

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

1        Autoencoder-based temporal models are routinely trained on a weighted sum of a  
2        reconstruction loss and a forecasting loss, with the weight treated as a hyperparam-  
3        eter. This practice assumes the balance is a genuine frontier. We ask whether it is  
4        instead a suboptimal operating point induced by forcing one latent to serve both ob-  
5        jectives. On a controlled chaotic benchmark (Lorenz-63 lifted to 30 noisy channels,  
6        known intrinsic dimension 3) we split the representation into an unbounded 3-d  
7        carrier read only by the forecaster and a sigmoid-bounded  $8 \times 8$  carrier read only  
8        by the decoder, trained in two stages so that the forecast gradient provably cannot  
9        reach the decoder. At matched parameters, matched wall clock and matched seeds,  
10       the split Pareto-dominates *every* setting of the weighted-loss knob that forecasts  
11       at all. We then report the result that complicates this: the advantage is compute-  
12       dependent — given  $10\times$  the wall clock, the balancing baseline overtakes it on both  
13       headline axes. What does not transfer with compute is stability: bounded decoding  
14       holds out-of-range rollouts at 0% against 100% for every unbounded baseline. The  
15       tradeoff is therefore not one fixed frontier, and compute-matched comparison alone  
16       cannot settle whether it is dissolvable.

## 17    1 Introduction

18    A large body of work trains one encoder to serve two objectives at once: reconstruct the present  
19    and forecast the future. The standard mechanism is a weighted loss,  $\mathcal{L} = (1 - \phi)\mathcal{L}_{\text{rec}} + \phi\mathcal{L}_{\text{fcst}}$ ,  
20    with  $\phi$  tuned per dataset. Coupled Attention Networks [Coupled Attention, 2023] call their weights  
21    “a tradeoff between the prediction and reconstruction models”; CAAE [CAAE, 2026] names the  
22    “objective mismatch between training for reconstruction fidelity and [the downstream task]”; a  
23    cluster of reconstruction-plus-prediction hybrids follows the same recipe [MemAAE, 2021, HyVAE,  
24    2023]; and SSP [Anonymous, 2026] pursues the same separation by spectral truncation, “decoupling  
25    reconstruction fidelity from rollout regularity.” Across all of them the tradeoff is something to *tune*.  
26    Almost nobody asks whether the tuning is necessary.

27    **What we are not claiming.** Rate–distortion theory [Shannon, 1959, Berger, 1971] proves a frontier  
28    between rate and distortion for a source, and the information bottleneck [Tishby et al., 2000] extends  
29    this to predictive representations. We are not claiming to beat any such bound. Rate–distortion  
30    constrains distortion *at a fixed rate*; it says nothing about which point inside the achievable region a  
31    given architecture and schedule lands on. Our claim is only the latter: at fixed capacity and compute,  
32    a shared latent under a weighted loss does not reach the best available point, and separating the  
33    carriers moves it.

34 **Contributions.** (i) A controlled, compute-matched test of whether the tradeoff is a frontier or  
 35 an operating point, on a system of known intrinsic dimension. (ii) A bounded/unbounded split  
 36 Pareto-dominating every weighted-loss setting that forecasts at all, at matched parameters, wall clock  
 37 and seeds. (iii) A negative result constraining this: the dominance does not survive a  $10\times$  baseline  
 38 budget. (iv) A bounded-shared-latent control isolating which ingredient buys what.

## 39 2 Method

40 **Objects.** For  $x_t \in \mathbb{R}^n$  write  $h_t = E(x_t) \in \mathbb{R}^n$ ,  $b_t = f(h_t) \in \mathbb{R}^k$  and  $C_t = m(b_t) \in [0, 1]^{a \times b}$ ,  
 41 with a decoder  $D : [0, 1]^{a \times b} \rightarrow \mathbb{R}^n$ , a forecaster  $g : \mathbb{R}^k \rightarrow \mathbb{R}^k$ , and a teacher  $r : \mathbb{R}^n \rightarrow [0, 1]^{a \times b}$ .  
 42 Here  $b$  is the *unbounded* carrier ( $k = 3$ ) and  $C$  the *bounded* one ( $a = b = 8$ ). Reconstruction runs  
 43  $x_t \rightarrow h_t \rightarrow b_t \rightarrow C_t \rightarrow \hat{x}_t$ ; forecasting steps only  $b$  and recomputes  $C$  from it:  $b_{t+1} = g(b_t)$ ,  
 44  $C_{t+1} = m(b_{t+1})$ ,  $\hat{x}_{t+1} = D(C_{t+1})$ .

45 **Consistency.** The chain is coherent only if the indirect route to the bounded latent agrees with the  
 46 direct one,  $m(f(h_t)) \approx r(h_t)$ . As  $r$  sees all  $n$  dimensions and  $m$  only  $k$ , agreement forces  $f$  to retain  
 47 everything  $C$  needs while compressing  $30 \rightarrow 3$ .  $r$  is a training-only teacher, never run at inference.

48 **Two stages, and why the split is exact.** Stage 1 trains  $\theta_A = \{E, f, m, r, D\}$  on single timesteps:  
 49  $\mathcal{L}_A = \|D(m(f(E(x_t)))) - x_t\|^2 + \lambda \|m(f(E(x_t))) - r(E(x_t))\|_F^2$ . Stage 2 *freezes*  $\theta_A$ , runs the  
 50 frozen encoder over every training timestamp to obtain the series  $b_t$ , and trains  $g$  alone, free-running,  
 51 entirely in  $b$ -space:  $\mathcal{L}_B(\theta_g) = \frac{1}{H} \sum_{\tau=1}^H \|\hat{b}_{t+\tau} - b_{t+\tau}\|^2$  with  $\hat{b}_{t+\tau} = g^{\circ\tau}(b_t)$ . Neither  $m$  nor  $D$   
 52 appears in  $\mathcal{L}_B$ , so  $\partial\mathcal{L}_B/\partial\theta_A \equiv 0$ : improving the forecast *cannot* degrade reconstruction. This  
 53 is the decoupling, and it is an identity rather than a penalty. Contrast the weighted loss, where  
 54  $\partial\mathcal{L}_{\text{fcst}}/\partial\theta_{\text{enc}} \neq 0$  and  $\phi$  merely chooses which objective loses.

55 **Boundedness gives unconditional stability.**  $m$  ends in a sigmoid, so  $C_{t+\tau} \in [0, 1]^{a \times b}$  by con-  
 56 struction.  $D$  is continuous on that compact set, hence  $\|\hat{x}_{t+\tau}\| \leq \sup_C \|D(C)\| < \infty$  for all  $\tau$ ,  
 57 *independently of  $\hat{b}_{t+\tau}$* : however far the unbounded carrier drifts, the decoded trajectory cannot leave  
 58 a bounded set, so error saturates instead of exploding. This costs zero parameters.

## 59 3 Experimental setup

60 **Data.** Lorenz-63 [Lorenz, 1963] at  $dt = 0.01$ , lifted through a frozen random tanh map to  $n = 30$   
 61 channels with Gaussian noise at 5% of each channel’s spread. True intrinsic dimension is 3, so  $k$   
 62 is a probe, not a hyperparameter. Training pools 8 independent trajectories, each split 70/15/15  
 63 chronologically; 32 further trajectories are held out entirely. Every standardisation constant is fit on  
 64 the pooled training portion only, and the reconstruction floor is *measured* from the realised noise  
 65 draw (0.002492) rather than assumed as noise<sup>2</sup>.

66 **Baselines.** (B) weighted-loss AE+GRU,  $\phi \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1\}$ , 16-d shared latent; (D)  
 67 the same with its *shared* latent sigmoid-bounded through rollout — the control holding boundedness  
 68 fixed while removing the split; (C) a physics-informed AE whose 3-d latent is calibrated to true  
 69 Lorenz coordinates and rolled out by RK4 on the known equations, plus a misspecified  $\rho = 26$   
 70 variant; and persistence/climatology.

71 **Protocol.** Every model gets **124,482 trainable parameters and 120 s of wall clock on one pinned**  
 72 **CPU thread.** Ours, the PINN,  $\phi \in \{0.1, 0.25, 0.5\}$  and model D run 3 seeds; remaining  $\phi$  settings  
 73 run 1. We report reconstruction MSE over the measured floor; VPT (lead time at which normalised  
 74 error first crosses 0.4, in Lyapunov times [Vlachas et al., 2020]); error at 5 Lyapunov times; and  
 75 *divergence*, the fraction of rollouts leaving the training box.

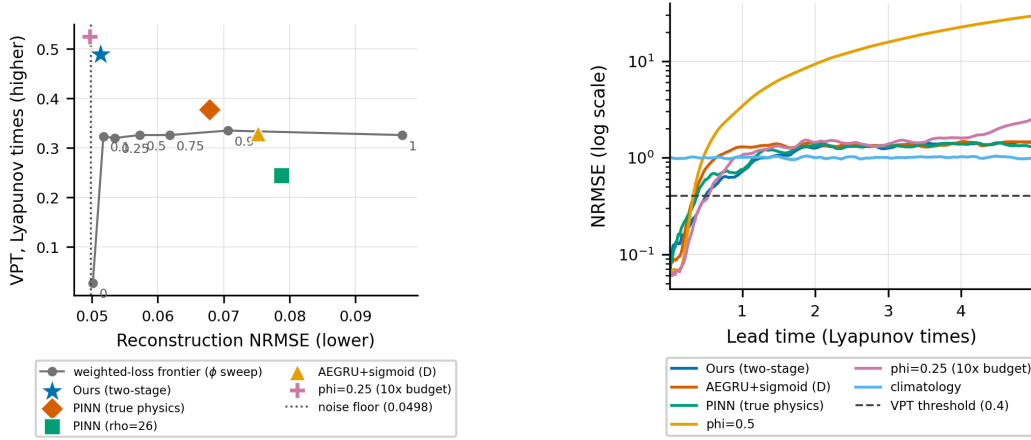


Figure 1: **Left:** the  $\phi$  frontier (grey) is nearly flat — paying reconstruction buys almost no forecast — and the split (star) sits outside the region it reaches. The  $10\times$ -budget point (pink) is the counterpoint: the frontier is not compute-invariant. **Right:** error vs lead time, log scale. Unbounded models climb without limit; every bounded decode path saturates at climatology ( $\approx 1$ ).

Table 1: Matched parameters, wall clock and seeds where marked.  $\times$ floor is reconstruction MSE over the measured floor (lower better); divergence is the fraction of rollouts leaving the training range. The  $10\times$  row is an off-protocol control, not a matched comparison. Full table in Appendix A.

| Model                           | seeds | $\times$ floor $\downarrow$ | VPT $\uparrow$                      | NRMSE@5LT $\downarrow$ | diverge $\downarrow$ |
|---------------------------------|-------|-----------------------------|-------------------------------------|------------------------|----------------------|
| <b>Ours (two-stage)</b>         | 3     | <b>1.06</b>                 | <b>0.489 <math>\pm</math> 0.013</b> | <b>1.18</b>            | <b>0%</b>            |
| $\phi=0$ (no forecasting)       | 1     | 1.01                        | 0.027                               | 30.6                   | 100%                 |
| $\phi=0.1$                      | 3     | 1.08                        | 0.323 $\pm$ 0.011                   | 15.5                   | 100%                 |
| $\phi=0.5$                      | 3     | 1.32                        | 0.326 $\pm$ 0.013                   | 15.8                   | 100%                 |
| AEGRU+sigmoid (D)               | 3     | 2.28                        | 0.329 $\pm$ 0.004                   | 1.28                   | 1.6%                 |
| PINN (true physics)             | 3     | 1.87                        | 0.378 $\pm$ 0.019                   | 1.19                   | 26%                  |
| persistence / climatology       | —     | —                           | 0.036 / 0.000                       | 1.37 / 1.00            | 0%                   |
| $\phi=0.25$ , $10\times$ budget | 1     | 0.99                        | 0.526                               | 1.43                   | 22%                  |

## 76 4 Results

77 **At matched compute, the split dominates.** The split reaches  $1.06\times$  the floor at VPT 0.489. Every  
78  $\phi$  that forecasts at all is worse on *both* axes at once: the best reaches 0.335 ( $\phi=0.9$ , App. A), and  
79 the only one reconstructing better ( $\phi=0$ ,  $1.01\times$ ) has VPT 0.027, below persistence — it does not  
80 forecast. The split is thus not another point on the knob’s curve but outside the region it reaches.  
81 That curve is itself nearly flat (0.320–0.335 across  $\phi \in [0.1, 1]$ ) while reconstruction degrades  $3.8\times$ :  
82 paying reconstruction buys almost no forecast here.

83 **The negative result: this is compute-dependent.** Held at 120 s the above reads as dissolution; it is  
84 not that simple. Given  $10\times$  the wall clock,  $\phi=0.25$  reaches VPT 0.526 at  $0.99\times$  the floor — better  
85 than the split on both headline axes. The shift ( $\Delta\text{VPT} = 0.205$ ) is an order of magnitude larger than  
86 the  $1\times$  run’s 3-seed spread ( $\sigma = 0.015$ ), so the family is *measurably undertrained* at that budget, not  
87 frontier-limited there. This bounds the claim: a compute-matched Pareto comparison, the standard  
88 protocol here, can invert under a compute rescaling. Any claim that architecture  $X$  dissolves the  
89 tradeoff, ours included, is only as strong as the budget it was measured at.

90 **What does not follow compute: stability.** Divergence separates the models in a way compute does  
91 not fix. Every unbounded baseline leaves the training range on 100% of rollouts at  $1\times$ ; at  $10\times$  that

falls only to 22%. The split is at 0%, with error at 5 Lyapunov times of 1.18 — the climatology level, i.e. saturating rather than exploding, as the boundedness argument predicts. That is a guarantee from the function class, not something better optimisation supplies.

**Which ingredient buys what.** Model D holds boundedness fixed and removes the split. It reaches divergence 1.6% — essentially the split’s stability — but reconstruction  $2.28\times$  floor and VPT 0.329, no better than the knob. The two ingredients buy different things: **boundedness buys stability; separating the carriers buys the reconstruction and forecast quality.** Neither alone reproduces the result — our cleanest evidence that the split does real work rather than acting as a reparameterised  $\phi$ .

**Schedule, noise, physics.** End-to-end training of the same modules degrades reconstruction  $4.4\times$  and VPT to 0.372: the schedule, not just the architecture, is load-bearing. At noise 0.15/0.30 (1 seed) the split still leads the best  $\phi$  — VPT gaps  $+0.163, +0.290, +0.208$  — but *not monotonically*, peaking at 0.15; we report the sequence, not a direction. The PINN needed a repair to be a fair comparator: its decoder reads the uncalibrated latent, so a *relative* physics residual was satisfiable in physically wrong coordinates. Recalibrating each epoch moved its forecast loss  $17.6 \rightarrow 0.015$  and its state-recovery  $R^2$  from  $[0.64, 0.70, 0.10]$  to  $[0.996, 0.989, 0.981]$ ; post-fix it has the best 50-Lyapunov-time attractor fidelity here (Wasserstein 0.068 vs the split’s 0.170). Appendices B–D.

## 5 Limitations

**One system:** Lorenz-63, one lift; the cross-domain phenomenon claim motivating this work is untested here. **Compute-dependence unresolved:** we did not run  $10\times$  with 3 seeds, locate the crossover, or give the split  $10\times$ . **Fixed capacity:** the bounded carrier is a fixed  $8 \times 8$  grid, so the adaptive-precision mechanism motivating the broader hypothesis is untested. **Uneven seeds:** single-seed rows report  $\sigma = 0$  because  $n = 1$ . **A floor artifact:**  $\phi=0.25$  at  $10\times$  sits at  $0.99\times$  the measured floor; a capacity-matched control (latent=3) clears it, indicating a wide unregularised latent exploiting cross-channel redundancy rather than a leak. **Constructed baselines:** the  $\phi$  sweep and model D are the balancing recipe in its plainest form, not named published methods.

## 6 Conclusion

At matched parameters, wall clock and seeds, separating a bounded decoder-facing carrier from an unbounded forecaster-facing one reaches an operating point no setting of the knob reaches, on a curve that is itself nearly flat. But that advantage is compute-dependent: we show it inverting at  $10\times$  budget. What does not depend on compute is stability — bounded decoding gives 0% divergence by construction where every unbounded baseline diverges on every rollout. The honest conclusion is narrower than “the tradeoff dissolves”: it is not a single fixed frontier, compute-matched comparison alone cannot settle whether it is dissolvable, and the reliability half of the problem is separable from the accuracy half.

## References

- O. Azencot, N. B. Erichson, V. Lin, and M. Mahoney. Forecasting sequential data using consistent Koopman autoencoders. In *ICML*, 2020.
- T. Berger. *Rate Distortion Theory: A Mathematical Basis for Data Compression*. Prentice-Hall, 1971.
- C. Durkan, A. Bekasov, I. Murray, and G. Papamakarios. Neural spline flows. In *NeurIPS*, 2019.
- D. Lee, C. Kim, S. Kim, M. Cho, and W.-S. Han. Autoregressive image generation using residual quantization. In *CVPR*, 2022.

134 Y. Liu, C. Li, J. Wang, and M. Long. Koopa: Learning non-stationary time series dynamics with  
135 Koopman predictors. In *NeurIPS*, 2023.

136 E. N. Lorenz. Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20(2):130–141,  
137 1963.

138 F. Mentzer, D. Minnen, E. Agustsson, and M. Tschannen. Finite scalar quantization: VQ-VAE made  
139 simple. In *ICLR*, 2024.

140 C. E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE Nat. Conv. Rec.*,  
141 4(142-163), 1959.

142 N. Tishby, F. C. Pereira, and W. Bialek. The information bottleneck method. *arXiv:physics/0004057*,  
143 2000.

144 P. R. Vlachas, J. Pathak, B. R. Hunt, T. P. Sapsis, M. Girvan, E. Ott, and P. Koumoutsakos. Back-  
145 propagation algorithms and reservoir computing in recurrent neural networks for the forecasting of  
146 complex spatiotemporal dynamics. *Neural Networks*, 126:191–217, 2020.

147 Stable long-horizon PDE forecasting via latent structured spectral propagators. *arXiv:2605.10154*,  
148 2026.

149 Contrastive adversarial autoencoder for multivariate time series. *Pattern Recognition*, 2026.

150 Coupled attention networks for multivariate time series anomaly detection. 2023.

151 Memory-augmented adversarial autoencoders for multivariate time-series anomaly detection. 2021.

152 Hybrid variational autoencoder for time series forecasting. 2023.

## 153 A Full results

Table 2: Complete comparison, all 14 configurations. Epochs are seed-0 counts within the 120 s budget (Ours reports stage 1 + stage 2 combined).

| Model                      | seeds | epochs | recon MSE | NRMSE  | VPT ( $\pm\sigma$ ) | diverge |
|----------------------------|-------|--------|-----------|--------|---------------------|---------|
| $\phi=0.25$ ( $10\times$ ) | 1     | 19     | 0.0025    | 0.0497 | 0.526               | 0.219   |
| Ours (two-stage)           | 3     | 16     | 0.0026    | 0.0513 | $0.489 \pm 0.013$   | 0.000   |
| PINN (true physics)        | 3     | 16     | 0.0047    | 0.0678 | $0.378 \pm 0.019$   | 0.260   |
| $\phi=0.9$                 | 1     | 2      | 0.0050    | 0.0706 | 0.335               | 1.000   |
| AEGRU+sigmoid (D)          | 3     | 2      | 0.0057    | 0.0752 | $0.329 \pm 0.004$   | 0.016   |
| $\phi=0.75$                | 1     | 2      | 0.0038    | 0.0618 | 0.326               | 1.000   |
| $\phi=1$                   | 1     | 2      | 0.0095    | 0.0971 | 0.326               | 1.000   |
| $\phi=0.5$                 | 3     | 2      | 0.0033    | 0.0572 | $0.326 \pm 0.013$   | 1.000   |
| $\phi=0.1$                 | 3     | 2      | 0.0027    | 0.0517 | $0.323 \pm 0.011$   | 1.000   |
| $\phi=0.25$                | 3     | 2      | 0.0029    | 0.0535 | $0.320 \pm 0.015$   | 1.000   |
| PINN ( $\rho=26$ )         | 1     | 16     | 0.0062    | 0.0788 | 0.245               | 0.000   |
| persistence                | —     | —      | —         | —      | 0.036               | 0.000   |
| $\phi=0$                   | 1     | 2      | 0.0025    | 0.0501 | 0.027               | 1.000   |
| climatology                | —     | —      | —         | —      | 0.000               | 0.000   |

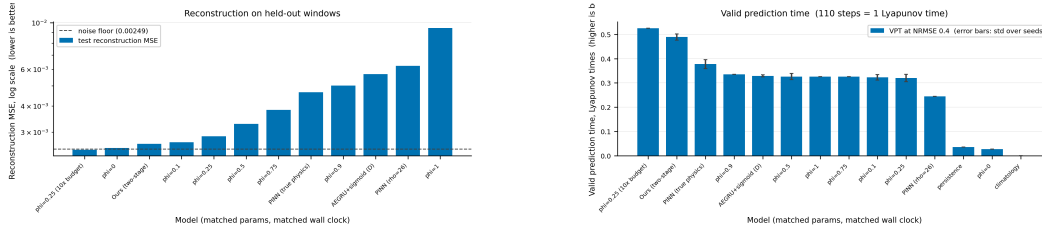


Figure 2: The two headline axes as bar charts. Left: reconstruction against the measured noise floor (dashed). Right: valid prediction time, error bars are the standard deviation over seeds — bars without one are  $n = 1$ .

154 Note the epochs column: the  $\phi$  models complete  $\sim 2$  epochs against Ours' 16 in the same wall clock,  
155 because the joint model re-encodes full sequences every step while stage 1 trains on single timesteps  
156 and stage 2 on precomputed latents. Equal seconds is not equal gradient steps; this is the mechanism  
157 behind the budget result of §4, and it is a property of the architectures, not a protocol error.

## 158 B Long-horizon attractor statistics (50 Lyapunov times)

Table 3: Attractor statistics after 5,500 free-running steps. Wasserstein is the mean 1-D distance between per-channel marginals of the rollout and of the truth (lower better).

| Model               | Wasserstein ↓ | spectrum log-err ↓ | diverge | final NRMSE |
|---------------------|---------------|--------------------|---------|-------------|
| PINN (true physics) | 0.068         | 1.88               | 0.906   | 1.42        |
| PINN ( $\rho=26$ )  | 0.149         | 2.07               | 0.188   | 1.42        |
| Ours (two-stage)    | 0.170         | 2.06               | 0.000   | 1.44        |
| AEGRU+sigmoid (D)   | 0.568         | 3.14               | 0.000   | 1.46        |
| $\phi=1$            | 139.3         | 8.24               | 1.000   | 389.7       |
| $\phi=0.5$          | 170.1         | 8.32               | 1.000   | 417.9       |
| $\phi=0$            | 213.1         | 8.59               | 1.000   | 534.3       |

159 After 5,500 blind steps the split still traces the attractor (Wasserstein 0.170) while every weighted-loss  
 160 setting is off by three orders of magnitude. The repaired PINN is best on this axis, as expected of a  
 161 model handed the exact equations.

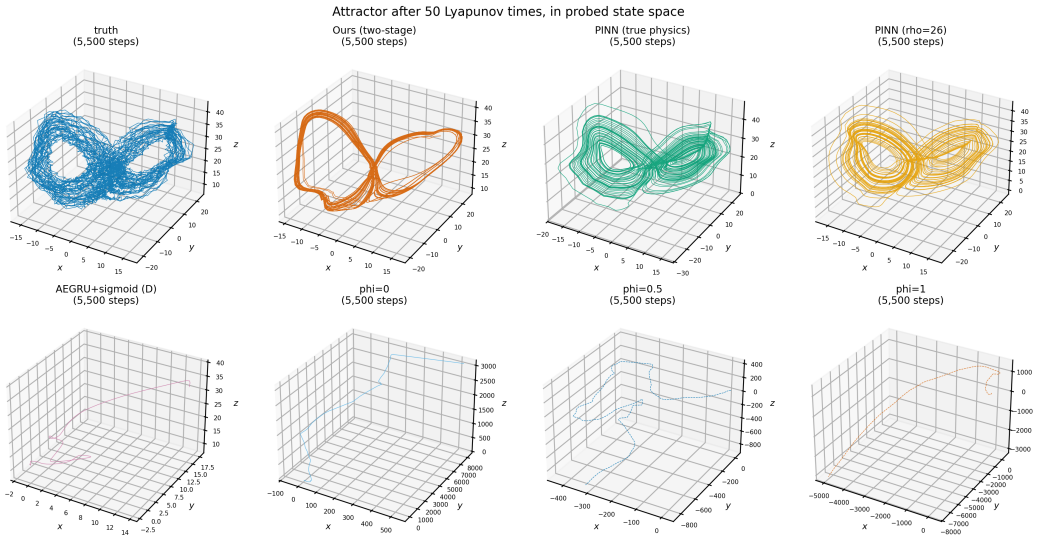


Figure 3: Attractor after 50 Lyapunov times (5,500 free-running steps), in probed state space. Truth first. The split and both PINNs still trace a recognisable butterfly; the weighted-loss settings have left the attractor entirely.

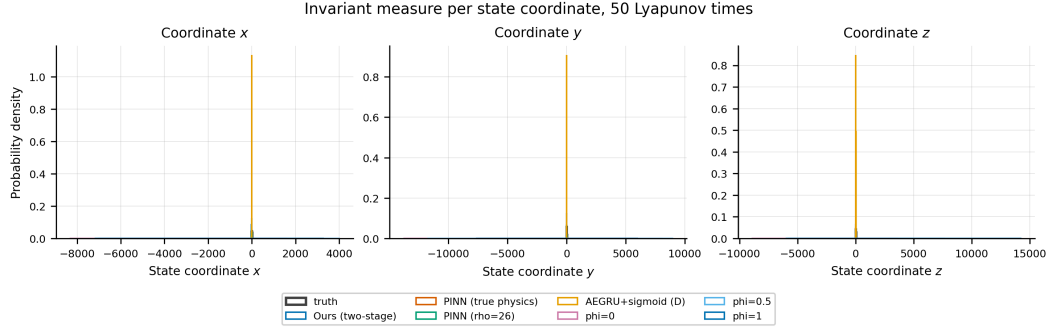


Figure 4: Invariant measure per state coordinate at 50 Lyapunov times — the distribution the Wasserstein column scores. Curves that collapse to a spike, or sit far off truth, correspond to rollouts that have diverged.

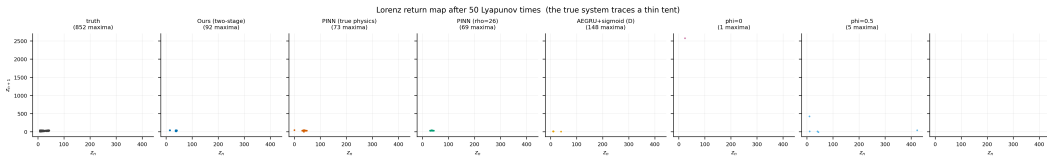


Figure 5: Lorenz return map (successive maxima of  $z$ ) after 50 Lyapunov times. The true system traces a thin tent; a model with the right climate but the wrong geometry produces a visibly fattened or collapsed one.

## 162 C Noise sweep

Table 4: Headline comparison re-run at three observation-noise levels; one seed at 0.15 and 0.30.

| noise | model                 | $\times$ floor | VPT           | diverge       |
|-------|-----------------------|----------------|---------------|---------------|
| 0.05  | Ours                  | 1.06           | 0.489         | 0.000         |
| 0.05  | $\phi=0.1 / \phi=0.5$ | 1.08 / 1.32    | 0.323 / 0.326 | 1.000         |
| 0.05  | AEGRU+sigmoid (D)     | 2.28           | 0.329         | 0.016         |
| 0.15  | Ours                  | 0.93           | 0.598         | 0.000         |
| 0.15  | $\phi=0.1 / \phi=0.5$ | 0.86 / 0.95    | 0.299 / 0.308 | 1.000         |
| 0.15  | AEGRU+sigmoid (D)     | 1.08           | 0.290         | 0.000         |
| 0.30  | Ours                  | 0.92           | 0.462         | 0.000         |
| 0.30  | $\phi=0.1 / \phi=0.5$ | 0.55 / 0.86    | 0.254 / 0.254 | 0.000 / 1.000 |
| 0.30  | AEGRU+sigmoid (D)     | 0.92           | 0.245         | 0.000         |

163 At noise 0.15 and 0.30 several models fall below  $1.0\times$  the measured floor. This is the same capacity  
 164 artifact discussed in the limitations, amplified: with more noise there is more to denoise, and cross-  
 165 channel redundancy lets a wide latent recover the clean signal slightly better than the per-channel  
 166 floor implies. It affects all models equally and does not change the ordering.

## 167 D Boundedness audit and training cost

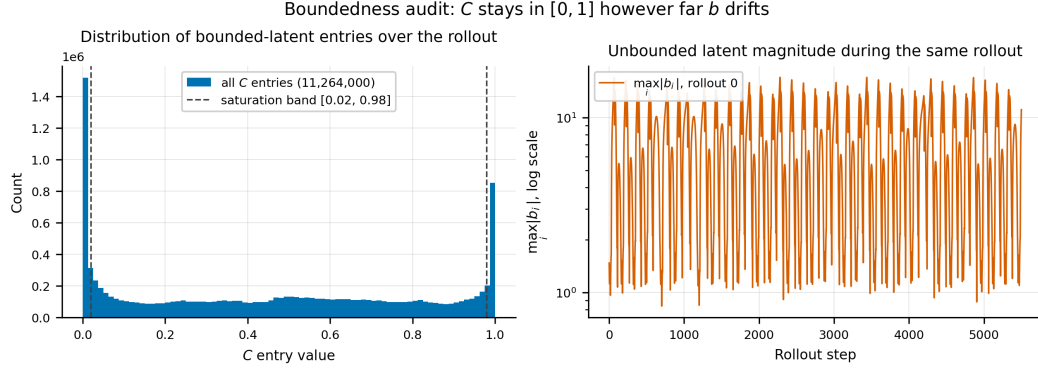


Figure 6: Left: distribution of bounded-latent entries over a 50-Lyapunov-time rollout — every value inside  $[0, 1]$ , as the construction guarantees. Right: the unbounded carrier’s magnitude over the same rollout. The bound holds on  $C$  regardless of what  $b$  does, which is the mechanism behind the 0% divergence in Table 1.

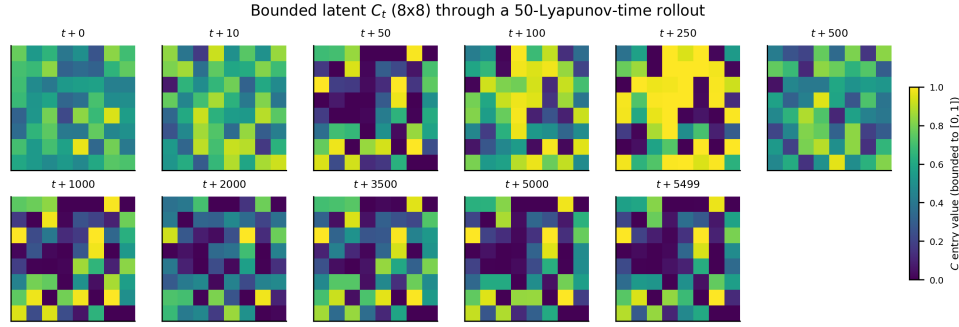


Figure 7: The bounded latent  $C_t$  ( $8 \times 8$ ) sampled through a 50-Lyapunov-time rollout. It neither saturates to the rails nor escapes the box.

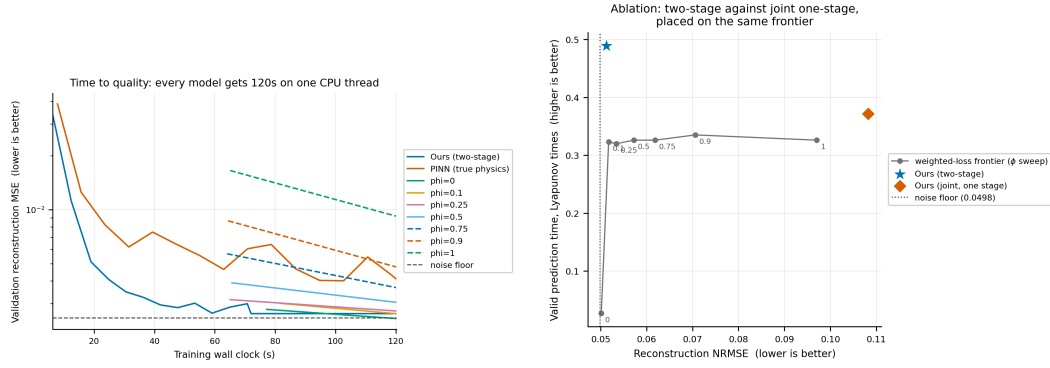


Figure 8: Left: validation reconstruction against wall clock — the split reaches  $2\times$  the noise floor in 24.8 s against 63–120 s for the baselines, because stage 2 trains on precomputed latents with no decoder in the graph. Right: the two-stage schedule against end-to-end training of the identical modules, placed on the same frontier.

## 168 **E Reproducibility**

169 Code, data generator and the executed notebook (all outputs stored) accompany this submission. All  
170 numbers come from a single end-to-end run at `QUICK=False` on one pinned CPU thread; the full  
171 run takes approximately three hours from an empty checkpoint directory. Every model is cached  
172 after training, so re-running reproduces the tables without retraining. Architecture:  $E$  is an MLP  
173  $30 \rightarrow 128 \rightarrow 128 \rightarrow 30$ ;  $f$  is  $30 \rightarrow 128 \rightarrow 64 \rightarrow 3$ ;  $m$  is  $3 \rightarrow 64 \rightarrow 128 \rightarrow 64$  with a sigmoid;  $r$  is  
174  $30 \rightarrow 128 \rightarrow 128 \rightarrow 64$  with a sigmoid;  $D$  is  $64 \rightarrow 128 \rightarrow 128 \rightarrow 30$ ;  $g$  is a residual GRU ( $3 \rightarrow 64$ ) with  
175 a linear head. All activations GELU. Adam,  $\eta = 10^{-3}$ ,  $\lambda = 1$ , stage-1 batch 256, stage-2 batch 64,  
176 warm-up 64 steps, training rollout 20 steps, evaluation horizon 550 steps (5 Lyapunov times).